



Introduction To Algorithms

Week 2 - Growth and Big-O Intuition:
"When Working Code Starts to Slow Down"

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Terms to Remember

Term	Brief Meaning
Performance analysis	Studying how an algorithm behaves as resources & input size change
Space complexity	How memory use changes as input grows
Time complexity	How the amount of work or time changes as input grows
Input size	The amount of data the algorithm processes
Big-O	A notation for describing growth behavior
Best case	The easiest or fastest likely situation for an algorithm
Worst case	The hardest or slowest likely situation for an algorithm
Average case	Expected behavior across typical inputs
Exact algorithm	Designed to produce an exact correct result
Approximate algorithm	Designed to produce a good-enough result under constraints
Randomized algorithm	Uses randomness as part of its process
Explainability	Ability to describe how or why an algorithm produced a result

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What We Will Use Today

- ▶ correct, understandable, efficient
- ▶ space and time analysis
- ▶ input size and growth behavior
- ▶ Big-O vocabulary
- ▶ best, worst, and average case
- ▶ timing evidence and its limits
- ▶ algorithm selection and explanation

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What We Will "Touch On" and Revisit Later

- Formal mathematical notation
- Recursion-based complexity
- Nested recursion
- Memory optimization
- GPU/CPU details
- Randomized and approximate algorithms in deeper AI/data contexts



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Review Coding Lab

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Review: What Lab 01 Taught Us

Last week, we focused on:

- precise rules
- expected results
- actual results
- edge cases
- revision based on evidence

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Today's Question

What happens when the amount of data gets larger?

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From “Works” To “Scales”

Correctness asks:

- ▶ Does the algorithm produce the expected result?

Growth asks:

- ▶ How does the work change as input grows?

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What Counts as Success Today

- compare two approaches
- collect timing evidence
- describe growth cautiously
- use basic Big-O vocabulary
- identify a limitation of your evidence

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Algorithm Quality

The reading describes strong algorithms as:

- ▶ Correct
- ▶ Understandable
- ▶ Efficient

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Performance Analysis

Performance analysis asks:

- ▶ How much time might this take?
- ▶ How much memory might this use?
- ▶ What happens as input size grows?

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Space And Time

Two common analysis types:

- ▶ Space analysis: memory growth
- ▶ Time analysis: work or runtime growth

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Space Complexity

Space complexity asks how memory use changes.

Patterns can be:

- ▶ Flat
- ▶ Converging
- ▶ Diverging



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Textbook Review 

Time Complexity

Time complexity asks:

- ▶ Can this scale?
- ▶ What work does the algorithm repeat?
- ▶ What happens with larger datasets?
- ▶ What resources are involved?



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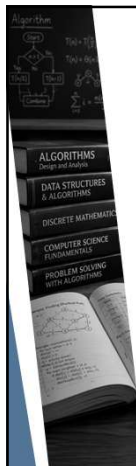
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Input Size

Input size is the amount of data the algorithm works on.

Examples:

- ▶ number of list items
- ▶ number of records
- ▶ number of characters
- ▶ number of graph nodes



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Performance Analysis Foundations



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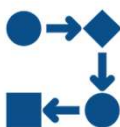


Growth Pattern

Growth pattern means:

- ▶ how work changes
- ▶ as input size changes

Small inputs can hide the pattern.



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Timing Is Evidence

Timing can show:

- ▶ what happened in a run
- ▶ on this machine
- ▶ with this code
- ▶ under these conditions



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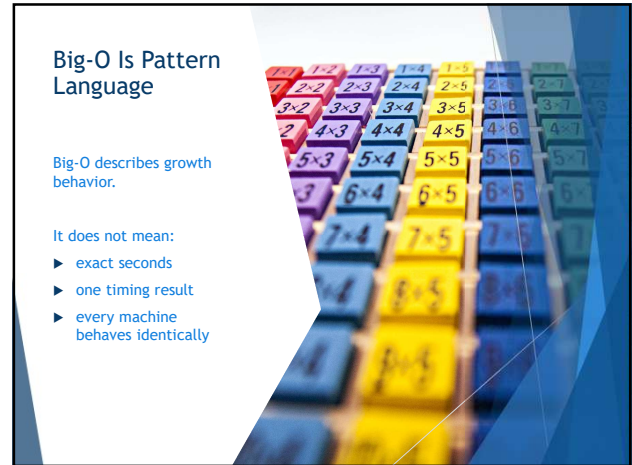
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Timing Can Be Noisy

Timing can vary because of:

- ▶ background activity
- ▶ hardware differences
- ▶ setup code
- ▶ number of trials
- ▶ input construction

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Big-O Is Pattern Language

Big-O describes growth behavior.

It does not mean:

- ▶ exact seconds
- ▶ one timing result
- ▶ every machine behaves identically

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What Counts As Work?

Work may include:

- ▶ comparisons
- ▶ loop iterations
- ▶ lookups
- ▶ calculations
- ▶ recursive calls
- ▶ memory writes

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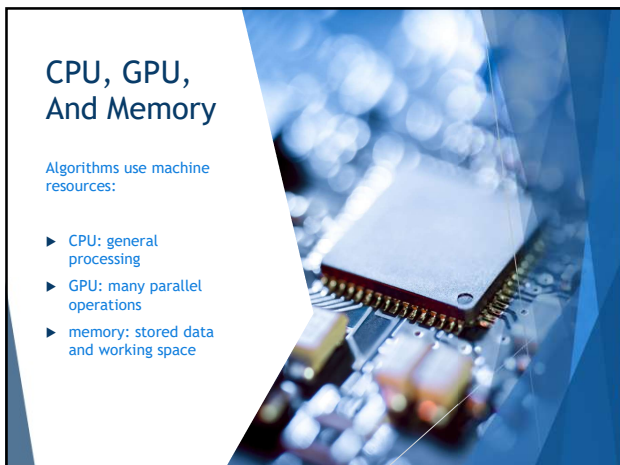
Space And Time Tradeoff

Sometimes an approach uses more memory to save time.

Example:

- ▶ list scanning: less structure, more repeated checking
- ▶ set lookup: extra structure, faster membership checks

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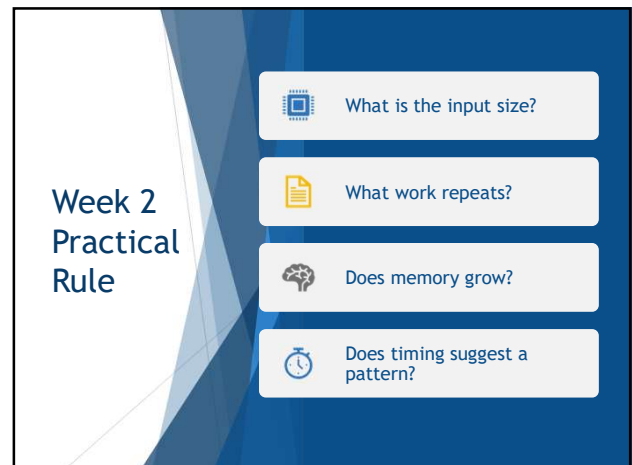


CPU, GPU, And Memory

Algorithms use machine resources:

- ▶ CPU: general processing
- ▶ GPU: many parallel operations
- ▶ memory: stored data and working space

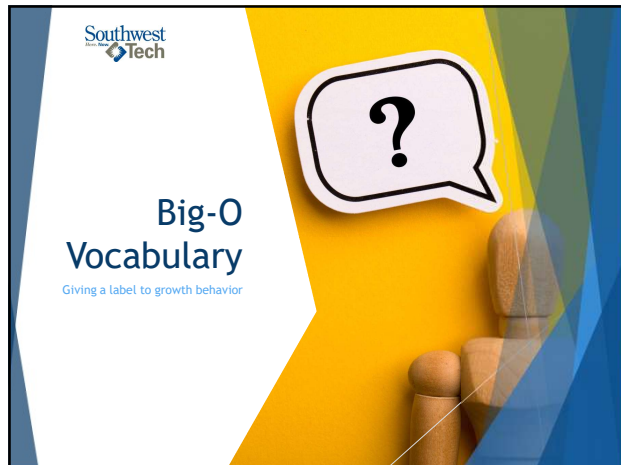
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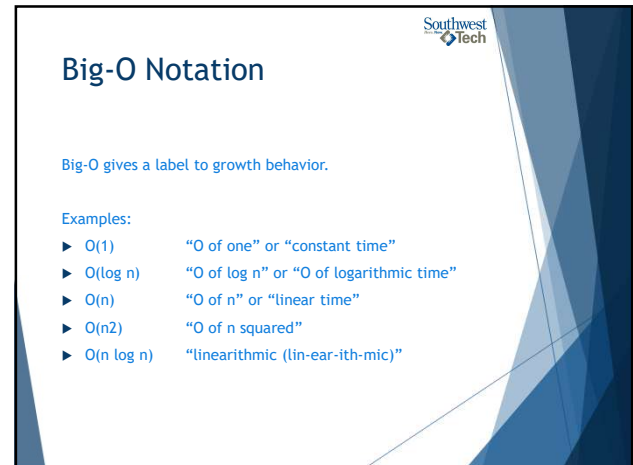
Week 2 Practical Rule

-  What is the input size?
-  What work repeats?
-  Does memory grow?
-  Does timing suggest a pattern?

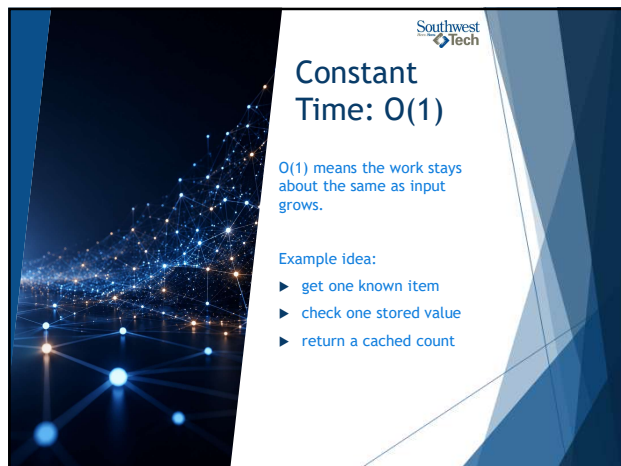
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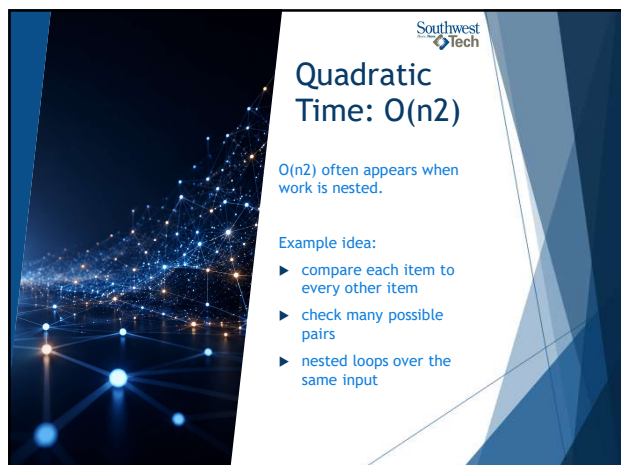
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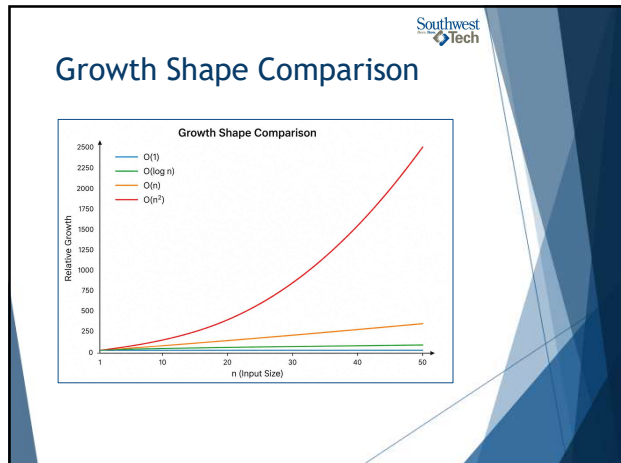
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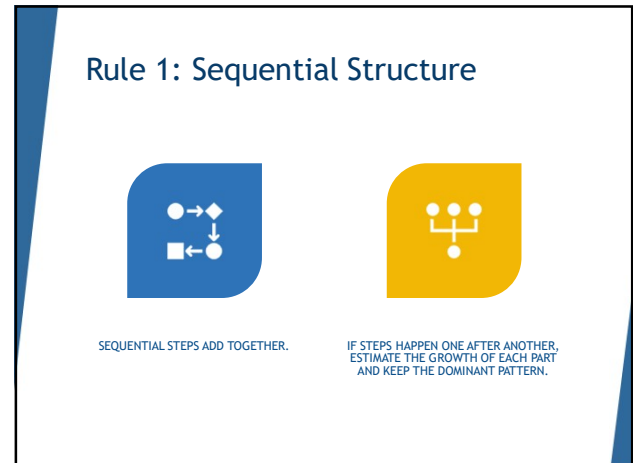
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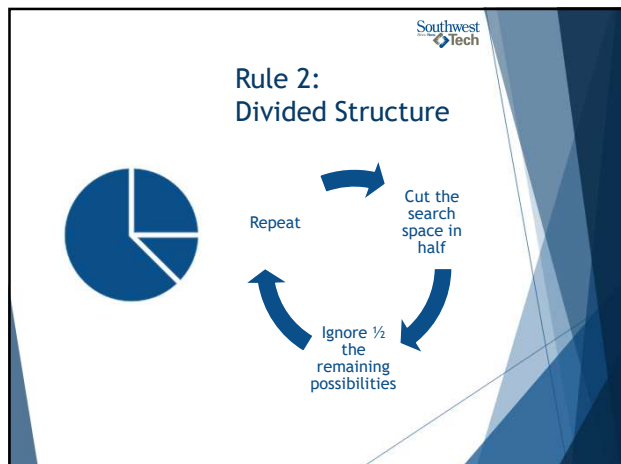
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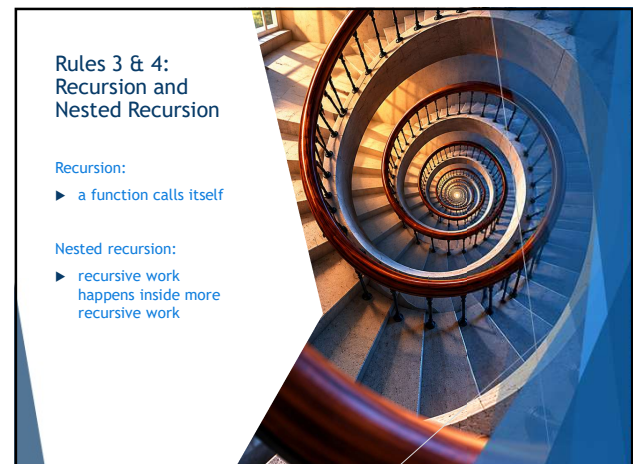
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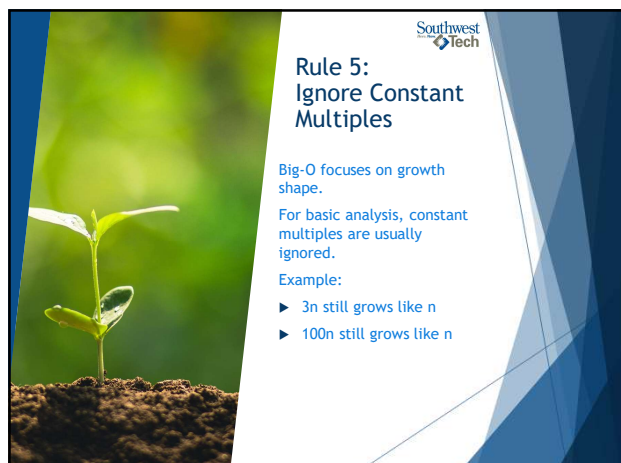
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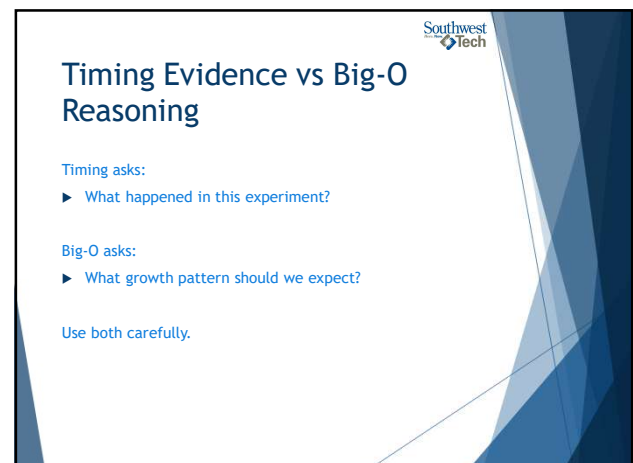
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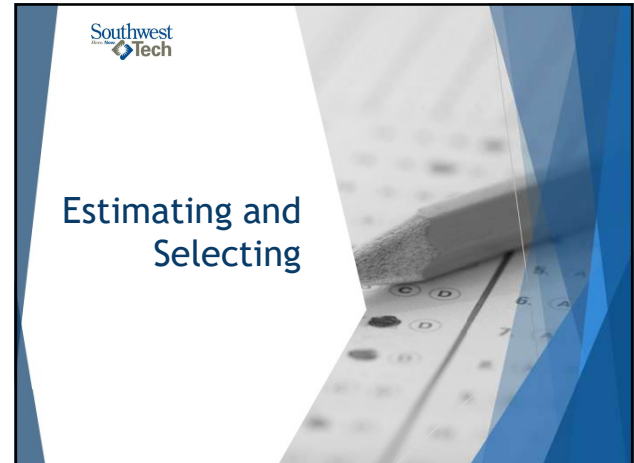
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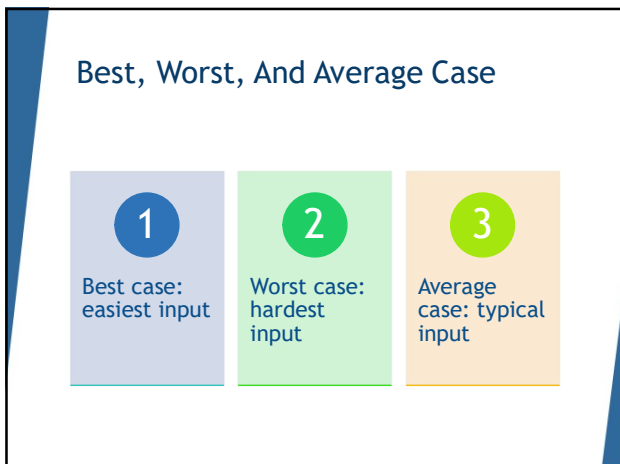
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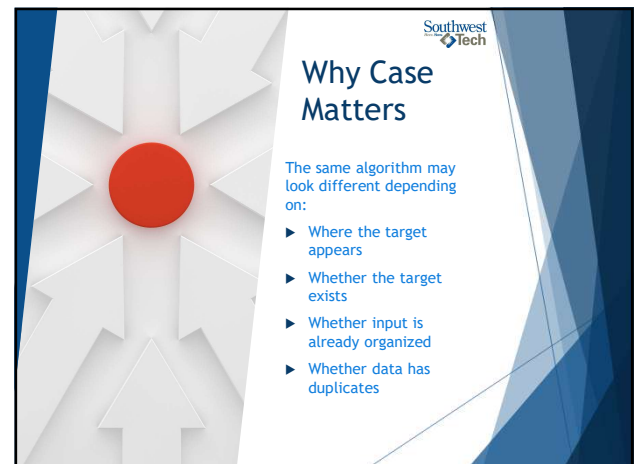
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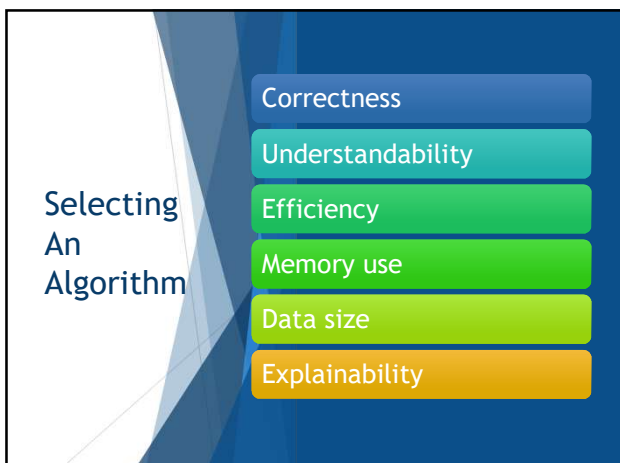
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What Evidence Does NOT Prove

A timing table does not prove:

- Every machine will match
- Every input will behave the same
- The algorithm is always best
- The code has no bugs

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Explainability

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Explainability

Explainability means you can describe:

- ▶ What the algorithm does
- ▶ Why the approach fits
- ▶ What evidence supports it
- ▶ What limits remain

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Good Growth Explanation

A responsible explanation says:

- ▶ What changed
- ▶ What evidence showed
- ▶ What Big-O label likely fits
- ▶ What the evidence does not prove



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Example

"In my timing table, Approach A increased more as input size grew. This matches the idea of repeated scanning, so linear or worse growth may be involved. However, my timing results are limited by machine noise and only four input sizes."



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Demo Scenario

Problem:

Compare two ways to check whether values appear in a collection:
 • Manual list lookup with a loop
 • Set membership lookup with 'in'
 The demo uses the same lookup task for both approaches, then repeats the task across larger input sizes.

Evidence:

1. A timing table across several input sizes.
 2. A formatted comparison summary that interprets the pattern.

Goal

Check whether values appear in a collection

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Predict The Work

List lookup may:

- ▶ Check many values
- ▶ Repeat scanning
- ▶ Grow with collection size

Set membership may:

- ▶ Use a structure built for lookup
- ▶ Avoid visible full scanning

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Demo Evidence

Watch the table for:

- ▶ Input size
- ▶ List lookup time
- ▶ Set lookup time
- ▶ Change across rows
- ▶ Reminder about noisy timing

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Demo Explanation

Demo explanation pattern:

- ▶ Both approaches were correct
- ▶ Timing changed as input grew
- ▶ List lookup grew more noticeably
- ▶ Set lookup changed less in this demo
- ▶ Exact numbers may vary

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AI-Assisted Complexity Review

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AI Can Help Review Complexity

AI can help you:

- ▶ Analyze likely time complexity
- ▶ Analyze likely space complexity
- ▶ Identify repeated work
- ▶ Explain Big-O in plain language
- ▶ Suggest alternative approaches



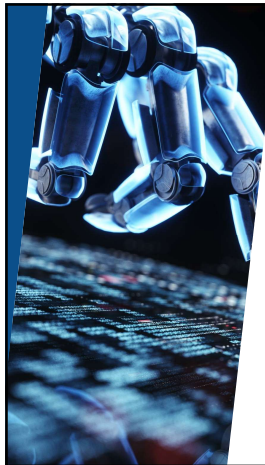
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Useful AI Prompt Pattern

“Analyze this Python function for likely time and space complexity.
Explain the reasoning in beginner-friendly language.
Identify repeated work.
Suggest one alternative if the current approach may not scale.
Do not rewrite the code unless I ask.”

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Verify The AI Review

After AI responds, check:

- ▶ Does it match the code?
- ▶ Does it identify the repeated work?
- ▶ Does it match the timing pattern?
- ▶ Did it overclaim?
- ▶ Can you restate it yourself?

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Demo Add-On: Console Color

After the plain demo works, add light console color.

The color helps highlight:

- ▶ section headings
- ▶ passed tests
- ▶ failed tests
- ▶ important boundary cases

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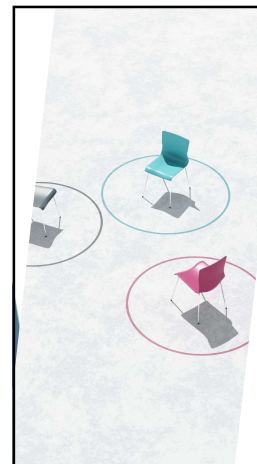
From Demo to Lab

In Lab 02, you will compare two approaches.

Your work must include:

- ▶ Two implementations
- ▶ At least four input sizes
- ▶ Timing table
- ▶ Chart or comparison table
- ▶ Growth explanation
- ▶ Limitation note

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Walkthrough Support

The walkthrough artifact can help you:

- ▶ Identify both approaches
- ▶ Predict repeated work
- ▶ Decide what to time
- ▶ Explain the difference

If you use it, you must explain the pattern in your own words, what you changes, and your decisions:

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README Evidence

Your README should show:

- ▶ approaches compared
- ▶ input sizes
- ▶ timing evidence
- ▶ chart or comparison table
- ▶ likely growth pattern
- ▶ timing limitation
- ▶ AI-use note, if applicable

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Wrap up

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What To Carry Forward



Performance analysis asks:

- ▶ Does it work?
- ▶ Is it understandable?
- ▶ How does it grow?
- ▶ What evidence supports the claim?

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Lab 02 Success Check



- ▶ Compares two correct approaches
- ▶ Uses increasing input sizes
- ▶ Reports measured timing
- ▶ Explains likely growth
- ▶ Names a limitation

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Next Step



For Lab 02:

- choose a comparison
- implement or attempt both approaches
- collect timing evidence
- build a table or chart
- explain the pattern cautiously

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How To Use The Textbook For Next Week's Reading



For next week, read for structure choice.

Focus on:

- ▶ What each structure is for
- ▶ What operations each structure makes easier
- ▶ What operations each structure makes harder
- ▶ How complexity tables can guide decisions
- ▶ Why tabular, matrix, and abstract structures appear in algorithm work

Do not memorize every operation cost. Use the tables like reference maps.

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Thank you!

Have Fun!

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